

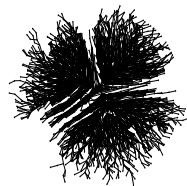
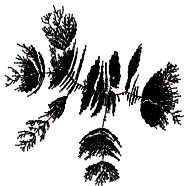
Descent-biased trees and dendrons

with Victor Dubach (Uppsala University) and Stephan Wagner (TU Graz)

Paul Thévenin

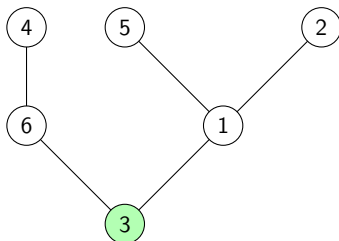
Discrete Random Structures II

May 19, 2026



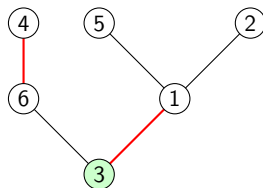
Trees

- Tree : connected acyclic graph.
- One marked vertex: the root. Oriented from the root towards the leaves.
- Vertices are labelled from 1 to n , where n is the number of vertices.



Descents

- $\mathcal{T}_n$ set of rooted trees with n labelled vertices (Cayley trees).
- An edge $e \in t$ is a *descent* if it goes from a vertex to another vertex with smaller label. $d(t)$ number of descents in a tree t .



$$d(t) = 2$$

Random descent-biased tree

- Fix $q \in [0, 1]$, and let $\mathcal{T}_n^{(q)} \in \mathfrak{T}_n$ be the random Cayley tree such that, for all $t \in \mathfrak{T}_n$:

$$\mathbb{P}\left(\mathcal{T}_n^{(q)} = t\right) = \frac{1}{Z_{n,q}} q^{d(t)},$$

where $Z_{n,q} = \sum_{t \in \mathfrak{T}_n} q^{d(t)}$.

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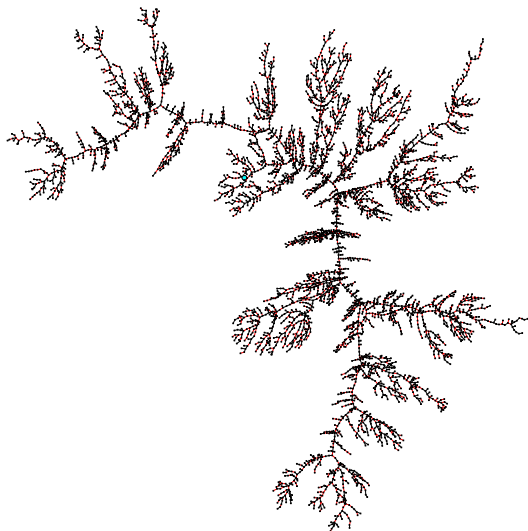
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- Interpolates between two well-known models: uniform Cayley trees ($q = 1$) and random recursive trees ($q = 0$).
- Goal: "study the structure" of $\mathcal{T}_n^{(q)}$ for n large.

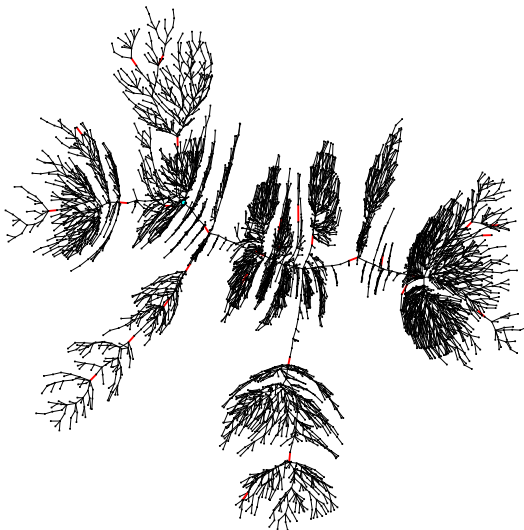
$n = 5000$, $q = 1$ (uniform Cayley tree)



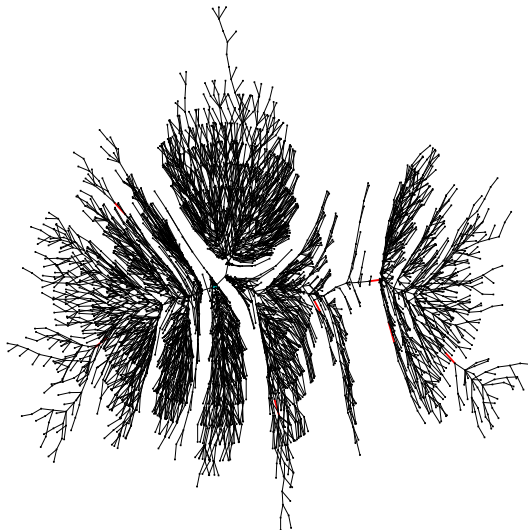
$$n = 5000, q = 0.5$$



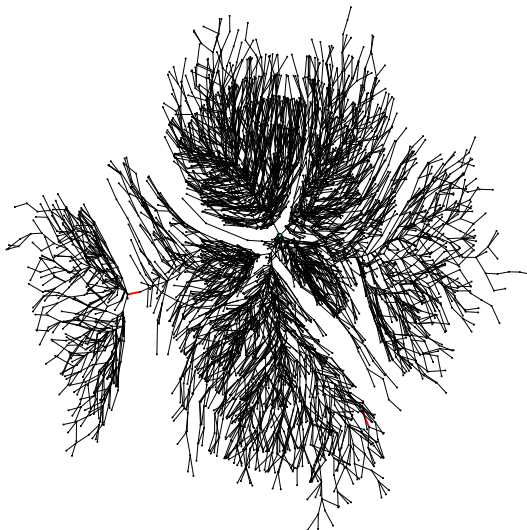
$$n = 5000, q = 1/1000$$



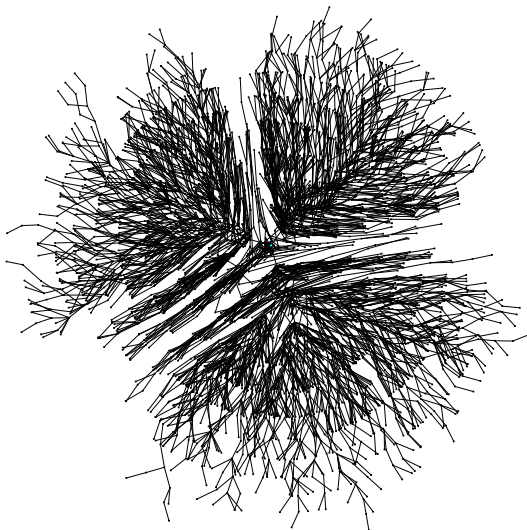
$$n = 5000, q = 1/2500$$



$$n = 5000, q = 1/10000$$



$n = 5000, q = 0$ (random recursive tree)



Brief history

- Descent-biased permutations [Euler 1755] (24153).

Eulerian polynomials: $A_n(q) = \sum_{\sigma \in \mathfrak{S}_n} q^{d(\sigma)}$.

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- *Local* limit of descent-biased permutations and trees [T. & Wagner '23]: what do finite neighbourhoods of the root look like?
- *Scaling* limit of descent-biased trees? [Dubach, T. & Wagner '26+]. Distances between uniform vertices.

Metric-measure space and convergence

Rooted m.m. space

A rooted metric-measure space (m.m. space) $(\mathcal{X}, \rho, d, \mu)$ is a complete separable metric space $(\mathcal{X}, d)$, with a marked point ρ and a probability Borel measure μ .

Metric-measure space and convergence

Rooted m.m. space

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We see a finite rooted tree (T, ρ) as a m.m. space $(T, \rho, d_T, \text{Unif}_T)$, where

- d_T is the usual graph distance;
- Unif_T the uniform measure on its vertices.

Metric-measure space and weak convergence

Weak convergence

Let $((\mathcal{X}_n, \rho_n, d_n, \mu_n))_{n \in \mathbb{N}}$ and $(\mathcal{X}_\infty, \rho_\infty, d_\infty, \mu_\infty)$ be rooted m.m. spaces, and $(a_n)_{n \geq 1}$ a sequence of positive numbers.

Metric-measure space and weak convergence

Weak convergence

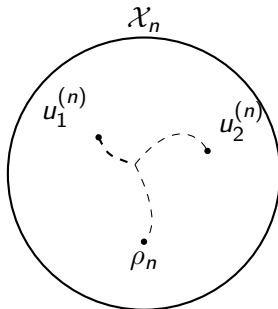
Let $((\mathcal{X}_n, \rho_n, d_n, \mu_n))_{n \in \mathbb{N}}$ and $(\mathcal{X}_\infty, \rho_\infty, d_\infty, \mu_\infty)$ be rooted m.m. spaces, and $(a_n)_{n \geq 1}$ a sequence of positive numbers.

We say that $a_n \mathcal{X}_n \xrightarrow[n \rightarrow \infty]{w} \mathcal{X}_\infty$ if, for all $p \geq 1$,

$$\left(a_n d_n(u_i^{(n)}, u_j^{(n)}) \right)_{0 \leq i, j \leq p} \xrightarrow[n \rightarrow \infty]{(d)} \left(d_\infty(u_i^{(\infty)}, u_j^{(\infty)}) \right)_{0 \leq i, j \leq p}.$$

where $(u_i^{(n)})_{1 \leq i \leq p}$ i.i.d. $\sim \mu_n$ in $\mathcal{X}_n$, and $u_0^{(n)} := \rho_n$.

For $p = 2$



$$\begin{pmatrix} 0 & d(\rho_n, u_1^{(n)}) & d(\rho_n, u_2^{(n)}) \\ d(\rho_n, u_1^{(n)}) & 0 & d(u_1^{(n)}, u_2^{(n)}) \\ d(\rho_n, u_2^{(n)}) & d(u_1^{(n)}, u_2^{(n)}) & 0 \end{pmatrix}$$

Weak convergence of trees

Weak convergence

Let $(\mathcal{T}_n, \rho_n)_{n \geq 1}$ be a sequence of (random) finite trees, $(\mathcal{X}_\infty, \rho_\infty, d_\infty, \mu_\infty)$ a rooted m.m. space, and $(a_n)_{n \geq 1}$ a sequence of positive numbers.

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where $(u_i^{(n)})_{1 \leq i \leq p}$ i.i.d. uniform in $\mathcal{T}_n$ and $u_0^{(n)} := \rho_n$, and $(u_i^{(\infty)})_{1 \leq i \leq p}$ i.i.d. $\sim \mu_\infty$ in $\mathcal{X}_\infty$ and $u_0^{(\infty)} := \rho_\infty$.

Case $q = 1$



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When $q = 1$, $\mathcal{T}_n^{(1)}$ is a uniform Cayley tree with n labelled vertices.

- We can count them : $|\mathcal{T}_n| = n^{n-1}$ [Borchardt 1860, Cayley 1889].
- Height of $\mathcal{T}_n^{(1)} = \Theta(\sqrt{n})$ [Rényi & Szekeres '65].
- Scaling limit: Brownian tree $(\mathcal{T}_\infty, \rho_\infty, d_\infty, \mu_\infty)$ [Aldous '91, '93], limit m.m. space.

The Brownian tree $\mathcal{T}_\infty$



Convergence for $q = 1$

Theorem [Aldous '91 '93], [Le Gall '05]

Let $(\mathcal{T}_n^{(1)}, \rho_n)$ be a uniform Cayley tree with n vertices. Let $(\mathcal{T}_\infty, \rho_\infty, d_\infty, \mu_\infty)$ be Aldous' Brownian tree. Let u_n, v_n be two i.i.d. uniform vertices in $\mathcal{T}_n^{(1)}$, and u, v two i.i.d. points of $\mathcal{T}_\infty$ with law μ_∞ . Then, in distribution, as $n \rightarrow \infty$:

$$\frac{1}{\sqrt{n}} d_n(u_n, v_n) \xrightarrow{(d)} d_\infty(u, v), \quad \frac{1}{\sqrt{n}} d_n(\rho_n, u_n) \xrightarrow{(d)} d_\infty(\rho_\infty, u).$$

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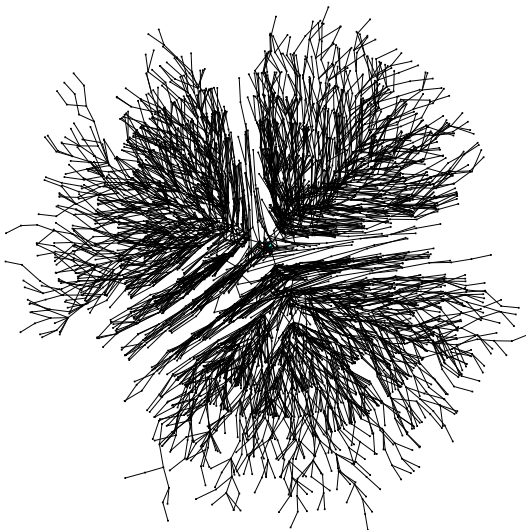
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True for several uniform i.i.d. vertices: for all fixed $p \geq 1$,

$$\frac{1}{\sqrt{n}} (d_n(u_n^{(i)}, u_n^{(j)}))_{1 \leq i, j \leq p} \xrightarrow[n \rightarrow \infty]{(d)} (d_\infty(u^{(i)}, u^{(j)}))_{1 \leq i, j \leq p}.$$

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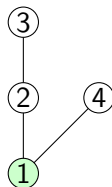
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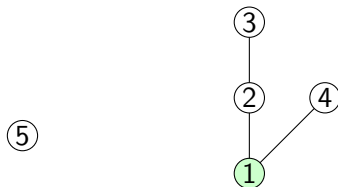
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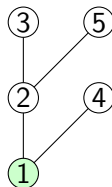


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$\mathcal{T}_5^{(0)}$



Some results

- We can count them: there are $(n - 1)!$ recursive trees on n vertices.
- Height of the tree $\mathcal{T}_n^{(0)} \sim e \log n$ [Pittel '94].
- Height of a typical vertex in $\mathcal{T}_n^{(0)} \sim \log n$.

"Scaling limit" for random recursive trees

- No nice metric space as a scaling limit for $\mathcal{T}_n^{(0)}$, even in the weak sense...

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Lemma

Let u_n, v_n two i.i.d. uniform vertices in $\mathcal{T}_n^{(0)}$, and ρ_n its root. Then we have:

$$\frac{1}{\log n} d_n(\rho_n, u_n) \xrightarrow[n \rightarrow \infty]{(\mathbb{P})} 1; \quad \frac{1}{\log n} d_n(u_n, v_n) \xrightarrow[n \rightarrow \infty]{(\mathbb{P})} 2.$$

"Scaling limit"

- One still has convergence of distance matrices:

$$\frac{1}{\log n} \left(d(u_i^{(n)}, u_j^{(n)}) \right)_{0 \leq i, j \leq 2} \xrightarrow{n \rightarrow \infty} \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 2 \\ 1 & 2 & 0 \end{pmatrix}.$$

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- $\mathcal{T}_n^{(0)}$ "converges" towards a degenerate m.m. space $(\mathcal{X}, d, \mu)$:
 $d(u, v) = 2$ almost surely, for u, v i.i.d. $\sim \mu$ points of $\mathcal{X}$.
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 $d(u, v) = 2$ almost surely, for u, v i.i.d. $\sim \mu$ points of $\mathcal{X}$.
 In particular, $\mathcal{X}$ would not even be separable.
- In order to define a nicer limiting space: **dendrons** [Elek & Tardos '22].

Dendrons

Idea [Elek & Tardos '22]: define a "natural" limit space for random recursive trees, sort of generalized tree.

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Definition [Elek & Tardos '22]

A *dendron* $D = (T, \rho, d, \nu)$ is a rooted tree (T, ρ, d) along with a probability measure ν on the set $A_D := T \times [0, \infty)$.

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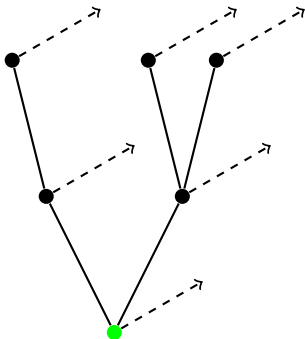
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We define a "pseudo-distance" function $d_D : A_D^2 \rightarrow \mathbb{R}_+$ by:

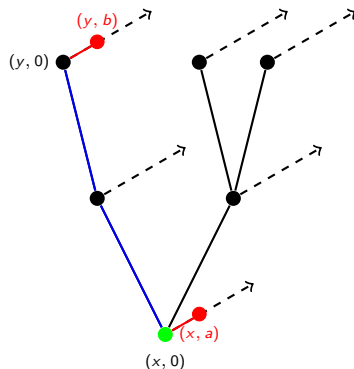
$\forall x, y \in T, \forall a, b \geq 0,$

$$d_D((x, a), (y, b)) := a + d(x, y) + b.$$

Dendron structure

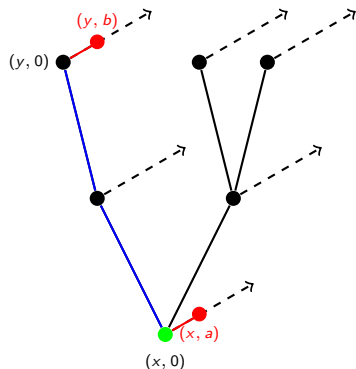


Dendron structure



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Dendron structure



$$d_D((x, a), (y, b)) = a + d(x, y) + b.$$

d_D is not a distance! $d_D((x, a), (x, a)) = a + a.$

Convergence towards dendrons

Definition [Elek & Tardos '22]

A sequence of (random) trees $(\mathcal{T}_n, \rho_n)$ converges towards a dendron (T, ρ, d, ν) if, for all $p \geq 1$:

$$\left(a_n d_{\mathcal{T}_n}(u_i^{(n)}, u_j^{(n)}) \right)_{0 \leq i, j \leq p} \xrightarrow[n \rightarrow \infty]{(d)} (d_D(u_i, u_j))_{0 \leq i, j \leq p}.$$

Here, $\{u_i^{(n)}, i \geq 1\}$ are i.i.d. $\sim \text{Unif}_{\mathcal{T}_n}$, $\{u_i, i \geq 1\}$ are i.i.d. $\sim \nu$, $u_0^{(n)} = \rho_n$ and $u_0 = \rho$.

Limits of trees and dendrons

Theorem [Elek & Tardos '22], [Janson '21]

- The space of dendrons is a Polish space.

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- Any sequence of random trees $(\mathcal{T}_n, \rho_n)_{n \geq 1}$ whose (renormalized) matrices of distances converge in distribution, converges towards a unique dendron.

Limits of trees and dendrons

Theorem [Elek & Tardos '22], [Janson '21]

- The space of dendrons is a Polish space.
- Any sequence of random trees $(\mathcal{T}_n, \rho_n)_{n \geq 1}$ whose (renormalized) matrices of distances converge in distribution, converges towards a unique dendron.
- Any dendron is the limit of a sequence of (renormalized) trees.

Dendron limit for $\mathcal{T}_n^{(1)}$



- Underlying tree $(T, d) = (\mathcal{T}_\infty, d_\infty)$.
- Mass measure $\nu = \mu_\infty \times \delta_0$, supported in $\mathcal{T}_\infty \times \{0\}$ (the half-lines have mass 0).

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- Let u, v i.i.d. $\sim \nu$. Then $u = (x, 0), v = (y, 0) \in \mathcal{T}_\infty$, and

$$d_D(u, v) = d_D((x, 0), (y, 0)) = 0 + d_\infty(x, y) + 0 = d_\infty(x, y).$$

- $\frac{1}{\sqrt{n}} \mathcal{T}_n^{(1)} \xrightarrow{w} \mathcal{T}_\infty$.

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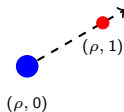


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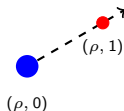
- $\frac{1}{\sqrt{n}} \mathcal{T}_n^{(1)} \xrightarrow{w} \mathcal{T}_\infty$.
- All trees are dendrons.

Dendron limit for $\mathcal{T}_n^{(0)}$



- Underlying tree $(T, d) = \{\rho\}$ (trivial distance).
- Mass measure $\nu = \delta_{(\rho, 1)}$.

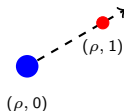
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- Let u, v i.i.d. $\sim \nu$. Then $u = v = (\rho, 1)$ and

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- Hence, $\mathcal{T}_n^{(0)} \xrightarrow[n \rightarrow \infty]{w} \Upsilon_{\delta_1}$.
- Notation $\Upsilon_\mu := \{\rho\} \times [0, \infty)$ with measure μ on $[0, \infty)$.

Other examples

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- Convergence of uniform trees, Bienaymé-Galton-Watson trees, towards Aldous' Brownian tree;
- Split trees, random recursive trees, some trees with logarithmic height, converge to Υ_{δ_1} [Janson '21];
- Uniform attachment trees with freezing converge (in some regime) towards Υ_{μ} for μ a measure with density [BBRKK '23].

Convergence of descent-biased trees

Theorem [Dubach, T. & Wagner '26+]

Let $(q_n)_{n \geq 1}$ be a sequence of elements in $[0, 1]$. Then we have (in the dendron convergence sense):

- (supercritical) If $q_n = q \in (0, 1]$, then $\frac{q-1}{\sqrt{q} \log q} \frac{1}{\sqrt{n}} \mathcal{T}_n^{(q)} \rightarrow \mathcal{T}_\infty$.

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- Interpolates between $\mathcal{T}_\infty$ ($a \rightarrow \infty$) and Υ_{δ_1} ($a \rightarrow 0$).

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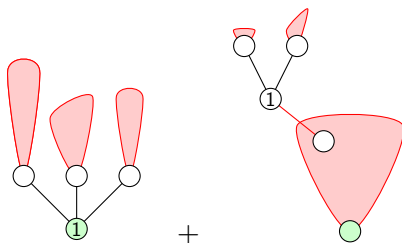
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- Interpolates between $\mathcal{T}_\infty$ ($a \rightarrow \infty$) and Υ_{δ_1} ($a \rightarrow 0$).
- Main tool: decomposition of the tree $\mathcal{T}_n^{(q_n)}$, generating function.

Tree decomposition

$$A(x, q) := \sum_t \frac{x^{|t|}}{|t|!} q^{d(t)}$$

Either the vertex labelled 1 is the root of the tree, or it is the endpoint of a descent.



Singularity analysis, typical height for q constant.

- This translates into:

$$\frac{\partial}{\partial x} A(x, q) = e^{A(x, q)} + q x e^{A(x, q)} \frac{\partial}{\partial x} A(x, q)$$

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- Height of a uniform vertex, for q constant:

$$\frac{q-1}{\sqrt{q} \log q} \frac{1}{\sqrt{n}} h(u_n) \xrightarrow[n \rightarrow \infty]{(d)} Z, \text{ independent of } q$$

Typical height for $q_n = a/n$

- Height of a uniform vertex in $\mathcal{T}_n^{(q_n)}$, for $q_n = a/n$:

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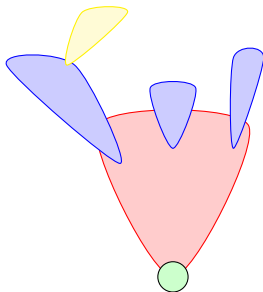
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- Hence, almost all vertices in $\mathcal{T}_n^{(a/n)}$ have a height which is an integer multiple of $\log n$. Structure of the tree?

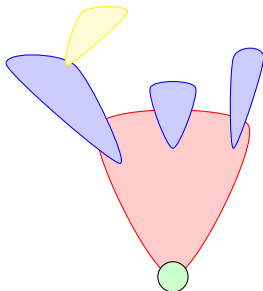
Structure of the tree

- The tree $\mathcal{T}_n^{(a/n)}$ consists in random recursive trees of macroscopic sizes $(c \cdot n)$, grafted on each other at uniform places.



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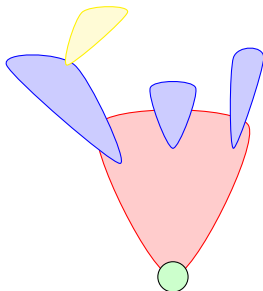
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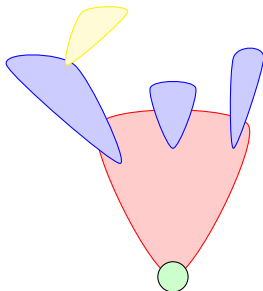
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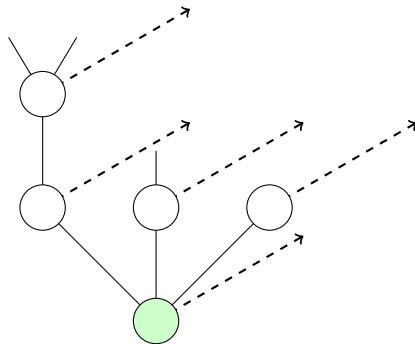
- Almost all vertices in the **red** part have height $\log n$;
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- Almost all vertices in the **yellow** part have height $3 \log n$.

Structure of the tree: $q_n = 1/n, n \gg 1$

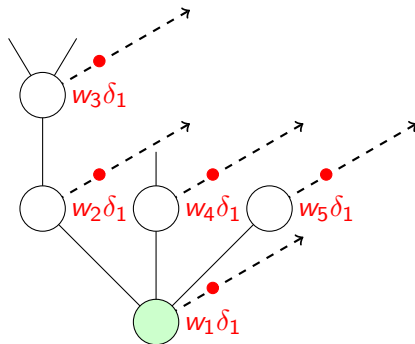


Components without descents

Structure of the dendron



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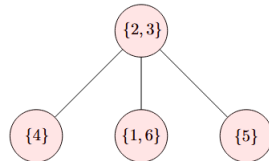
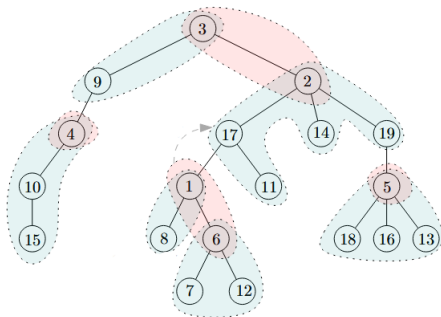
Edges have length 1.

$$\nu = \sum_{i \geq 1} w_i \delta_{(v_i, 1)}.$$

The weights $(w_i)_{i \geq 1}$ are random.

Additional results

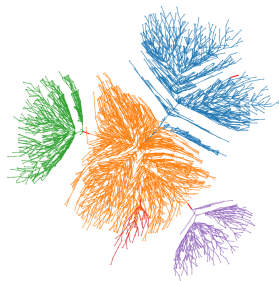
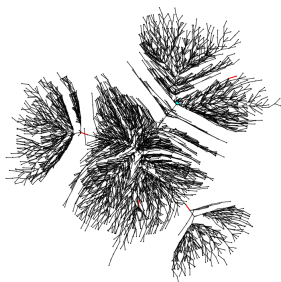
- We obtain the precise law of the limiting dendron, and a recursive construction.
- Proofs through a refined decomposition of the tree: look at the k smallest labels.



Conclusion

- Dendrons allow us to construct limiting spaces for trees with degenerate behaviours, as soon as distances between uniform vertices converge in distribution.
- Weak notion of convergence ; proving the convergence "boils down to" studying distances between finitely many points. Combinatorial methods are thus very suited (generating functions, singularity analysis, ...)

THANKS



The limiting dendron $\mathfrak{D}_a$

Construction of the base tree, which is an infinite tree with vertices $(v_i)_{i \geq 1}$.

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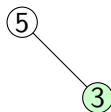
An example

$[3, 5, 2, 3, 1, 2, 5, 7, \dots]$

③

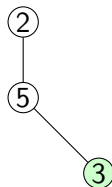
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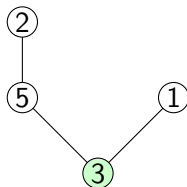
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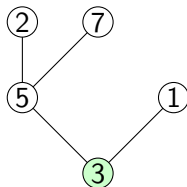
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Sketch of proof

- k -decomposition of the tree, consider only the first k labels.
- They are sorted in different small components, with descents inbetween.
- Reduced tree!
- One can compute the asymptotic probability of a reduced tree.
- For k large, the size of a component (and its degree in the reduced tree) are (asymptotically) proportional to the number of small labels in them.
- The reduced tree, conditionally on the degrees of the components, is a tree with a prescribed degree sequence.
- For k large, components are roughly random recursive trees - $\hookrightarrow$ we catch all the descents.

Reduced tree, $k = 6$, $n = 18$

